Adithya Cherian Abraham

Machine Learning Researcher · Deep Learning · NLP & Information Retrieval

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PROFILE

Final-year MCA student at Amrita Vishwa Vidyapeetham, Mysuru, with a strong foundation in deep learning, transformer architectures, and NLP. Experienced in designing and evaluating hybrid model pipelines spanning computer vision and information retrieval. Self-driven learner with a habit of reading arXiv pre-prints and translating mathematical ideas into working experiments. Seeking a research opportunity to contribute to and grow within a rigorous ML lab.

EDUCATION

Integrated Master of Computer Applications (MCA) · Amrita Vishwa Vidyapeetham, Mysuru

August 2022 – Present (Course Completion-2027)

- Specialisation: Machine Learning, Deep Learning, and Applied AI
- Coursework spans neural network mathematics, optimisation theory, and model architecture design

RESEARCH INTERESTS

- Transformer architectures and attention mechanisms — scaling laws, architectural innovations (attention residuals, MRL)
- Representation learning and dense retrieval — bi-encoders, cross-encoders, embedding optimisation
- Hybrid deep learning pipelines — feature extraction with transformers paired with classical ML classifiers
- Long-context memory modelling in large language models
- Generative Adversarial Networks (GANs) and their applications in adversarial simulation

PROJECTS

Cross-Lingual Information Retrieval (Hinglish → English) · Personal Research

2025

- Designed a bi-encoder retrieval system to fetch relevant English documents from Hinglish queries, addressing code-mixed language challenges
- Curated a training dataset with hard-negative and positive pairs; trained the bi-encoder using contrastive learning
- Integrated Matryoshka Representation Learning (MRL) to enrich per-dimension embedding quality and boost FAISS retrieval accuracy
- Experimented with cross-encoder reranking; analysed failure modes where the model failed to generalise on training pairs, deepening understanding of reranking pipeline design

Sugarcane Leaf Disease Detection — ViT + SVM Hybrid · Collaborative Faculty Research, Amrita Vishwa Vidyapeetham

November 2024 – April 2025

- Built a hybrid model using the feature extraction layers of a Vision Transformer (ViT) combined with an SVM fine-tuned using RBF kernel optimisation and hyperparameter search
- Achieved 96% test accuracy, outperforming KNN, Decision Tree, and Random Forest baselines
- Demonstrated how dimensionality reduction through deep feature extraction improves model focus on discriminative features

Smart Traffic Congestion Visualiser (STCV) · Personal Project

September 2025 – Present · github.com/cloud9ops/Smart-Traffic-Congestion-Visualizer-STCV-

- Built a traffic lane scheduling simulator using a custom starvation-aware priority heuristic (pLCF): Priority = Vehicles + (Starvation_Time × 0.25)
- Designed to prevent lane starvation while maintaining throughput under varying congestion scenarios

Grammar Correction Tool · Personal Project
June – July 2025 · github.com/cloud9ops/GrammarCorrectionTool

- Developed a modular offline grammar-correction system combining rule-based logic with ML techniques
- Built as an extensible NLP research platform suitable for academic writing and domain-specific style adaptation

SEEMA — Secure Smart Electricity Meter · IoT & Embedded Systems Project
July 2024 – June 2025

- Designed SEEMA, a low-cost IoT device for real-time electricity consumption monitoring with smart-grid integration
- Invented NRG Morph, a lightweight encryption algorithm tailored for resource-constrained IoT systems
- Patent filed — Application No. 202541043088 (Filed May 3, 2025; Status: Published)

Stock–War Correlation Analysis · Data Analysis Project
November 2024 · github.com/cloud9ops/Stocks-War-Comparison

- Analysed stock trends of arms, petroleum, and gold sectors during geopolitical events using news APIs and sentiment analysis
- Identified limitations of sentiment-driven prediction due to media bias, informing future work on robust signal extraction

TECHNICAL SKILLS

ML / DL Frameworks: PyTorch, scikit-learn, HuggingFace Transformers, FAISS, sentence-transformers
Architectures & Concepts: Transformers (ViT, BERT-family), SVMs, bi-encoders, cross-encoders, MRL, GANs, RAG
Languages: Python, C/C++, Java, C#, Bash, Lua
Tools & Platforms: Linux (Arch), Neovim, Git, MEAN Stack, LuaLaTeX

CERTIFICATIONS & COURSES

- Reinforcement Learning — Infosys Springboard (October 2023): Fundamentals of RL and its mathematical model
- Workshop on Generative AI — Amrita IIC Research Lab (September 2024): Embedding models, self-attention mechanisms, implemented a RAG chatbot
- Foundations of Cybersecurity — Google (November 2025): Part of the Google Cybersecurity Professional Certificate

CONFERENCES & MEMBERSHIPS

- TUG 2025 (TeX Users Group Conference), Thiruvananthapuram — July 2025: Explored advances in LaTeX tooling, tagpdf for accessible PDF generation, and CTAN ecosystem maintenance
- Member, TeX Users Group (July 2025 – Present)

HONOURS & AWARDS

- Second Place — Mind Meld Debate Competition, Amrita Kalaanveshane, Amrita Vishwa Vidyapeetham (November 2024)

LANGUAGES

Malayalam (Native)   **English** (C2 Proficient)   **French** (A1 Beginner)